Huawei Access Node KOR-ACC-29 Optical Transport Failure SOP

Node: KOR-ACC-29 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_3665

Description

The Huawei Access Node KOR-ACC-29 Optical Transport Failure is a critical issue that can occur due to various reasons such as SFP/XFP degradation, fiber cuts, Rx/Tx power loss, or hardware failures. This failure domain occurs on the Access role of the node due to its responsibility of handling the initial entry and exit points of the optical signals from the external network. The Optical Transport Network (OTN) protocol, which relies heavily on the SFP/XFP modules to transmit and receive optical signals, is responsible for the transmission of data over long distances. The OTN protocol uses a combination of Wavelength Division Multiplexing (WDM) and Time Division Multiplexing (TDM) to enable the transmission of multiple channels over a single fiber. In the event of a failure, the OTN protocol can fail to properly transmit or receive data, resulting in an optical transport failure. The Huawei Access Node KOR-ACC-29 is equipped with advanced features such as OTN switching, WDM, and TDM, making it an ideal platform for high-speed optical transport networks.

Pre-requisite Checks

- Ensure that the node is accessible and the power is on.
- Verify that no other maintenance is being performed on the node.
- Check the redundant link status to ensure that it is operational.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	Medium
Fiber cuts	High
Rx/Tx power loss	Low
Hardware failures	Medium

Diagnostic Commands & Expected Outputs

```
Command: display optical-transponder information, Expected Output: The output should display the SFP/XFP information, including the module type, status, and power consumption., Description: This command displays the SFP/XFP information, which can help identify any issues with the module.

Command: display optical-channel information, Expected Output: The output should display the OTN channel information, including the channel type, status, and power consumption., Description: This
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```
command displays the OTN channel information, which can help identify
any issues with the channel.

Command: display optical-fiber information, Expected Output: The
output should display the fiber information, including the fiber type,
status, and power consumption., Description: This command displays the
fiber information, which can help identify any issues with the fiber.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transponder , Description: Replace the SFP/XFP module with a new one to resolve the issue.

Step 2: Step: Repair the fiber cut, Command: repair optical-fiber , Description: Repair the fiber cut to resolve the issue.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust optical-channel power , Description: Adjust the Rx/Tx power to resolve the issue.

Post-Check Verification

```
Command: display optical-transponder information, Expected Output: The
output should display the SFP/XFP information, including the module
type, status, and power consumption., Description: This command
verifies that the SFP/XFP module is functioning correctly.

Command: display optical-channel information, Expected Output: The
output should display the OTN channel information, including the
channel type, status, and power consumption., Description: This
command verifies that the OTN channel is functioning correctly.

Command: display optical-fiber information, Expected Output: The
output should display the fiber information, including the fiber type,
status, and power consumption., Description: This command verifies
that the fiber is functioning correctly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore optical-transponder , Description: Restore the original SFP/XFP module to resolve the issue.

- Step: Restore the previous fiber configuration, Command: restore optical-fiber , Description: Restore the previous fiber configuration to resolve the issue.

- Step: Restore the original Rx/Tx power settings, Command: restore optical-channel power , Description: Restore the original Rx/Tx power settings to resolve the issue.